

- | | | | |
|---------|---------|---------|---------|
| 1. (C) | 2. (C) | 3. (B) | 4. (B) |
| 5. (C) | 6. (B) | 7. (B) | 8. (C) |
| 9. (A) | 10. (A) | 11. (D) | 12. (C) |
| 13. (D) | 14. (C) | 15. (D) | 16. (D) |
| 17. (B) | 18. (B) | 19. (B) | |

解析：

$$1. 240^\circ = \frac{4\pi}{3}$$

$$\widehat{AB} = 100 \times \frac{4\pi}{3} = \frac{400\pi}{3}$$

$$\text{一趟為 } 2\widehat{AB} = 2 \times \frac{400\pi}{3} = \frac{800\pi}{3}$$

又8公里=8000公尺

$$\text{依題意, } \frac{800\pi}{3} \times x > 8000 \Rightarrow x > \frac{30}{\pi} \approx 9.55$$

因此, x 最小為10

故選(C)

$$2. \because \theta = 3.1 \approx 3.1 \times 57.3^\circ = 177.63^\circ$$

$\therefore \theta$ 為第二象限角

故選(C)

3. 三角函數平方和關係式：

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\Rightarrow \tan^2 \theta - \sec^2 \theta = -1$$

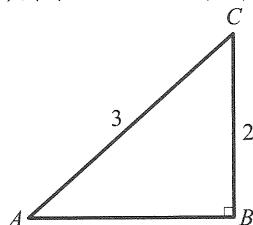
$$\Rightarrow (\tan \theta + \sec \theta)(\tan \theta - \sec \theta) = -1$$

$$\because \tan \theta + \sec \theta = 5$$

$$\therefore 5 \times (\tan \theta - \sec \theta) = -1$$

$$\Rightarrow \tan \theta - \sec \theta = -\frac{1}{5}$$

4. 如圖, $\triangle ABC$ 為直角三角形



由畢氏定理

$$\overline{AC}^2 = \overline{AB}^2 + \overline{BC}^2$$

$$\Rightarrow 3^2 = \overline{AB}^2 + 2^2 \Rightarrow \overline{AB}^2 = 5$$

$$\Rightarrow \overline{AB} = \sqrt{5}$$

$$\text{由三角函數的定義, } \tan A = \frac{\overline{BC}}{\overline{AB}} = \frac{2}{\sqrt{5}}$$

故選(B)

$$5. \sin 2024^\circ = \sin(360^\circ \times 5 + 224^\circ) = \sin 224^\circ$$

$$\because \sin \theta = \sin 224^\circ \text{ 且 } 180^\circ < \theta < 270^\circ$$

$$\therefore \theta = 224^\circ$$

$$6. \sin \theta \tan \theta = \sin \theta \times \frac{\sin \theta}{\cos \theta} = \frac{\sin^2 \theta}{\cos \theta}$$

$$\cos \theta \cot \theta = \cos \theta \times \frac{\cos \theta}{\sin \theta} = \frac{\cos^2 \theta}{\sin \theta}$$

$$\because \sin \theta \tan \theta < 0 \text{ 且 } \cos \theta \cot \theta > 0$$

$$\therefore \frac{\sin^2 \theta}{\cos \theta} < 0 \text{ 且 } \frac{\cos^2 \theta}{\sin \theta} > 0$$

而 $\sin^2 \theta$ 與 $\cos^2 \theta$ 為正數

則 $\cos \theta < 0$ 且 $\sin \theta > 0$

(1) 若 $\cos \theta < 0$, 則 θ 為第二象限角或第三象限角

(2) 若 $\sin \theta > 0$, 則 θ 為第一象限角或第二象限角

由(1)和(2)可知, 故 θ 為第二象限角

$$7. \because 2025^\circ = 360^\circ \times 5 + 225^\circ$$

$$\therefore \tan 2025^\circ = \tan 225^\circ,$$

$$\sec 2025^\circ = \sec 225^\circ$$

$$\text{而 } \tan 225^\circ = \tan(180^\circ + 45^\circ) = \tan 45^\circ = 1$$

$$\sec 225^\circ = \sec(180^\circ + 45^\circ) = -\sec 45^\circ$$

$$= -\frac{1}{\cos 45^\circ} = -\sqrt{2}$$

$$\text{故 } \tan 2025^\circ + \sec 2025^\circ = \tan 225^\circ + \sec 225^\circ$$

$$= 1 + (-\sqrt{2}) = 1 - \sqrt{2}$$

$$8. (1) \because y = |\sin x| \text{ 的週期為 } \pi$$

$$\therefore y = |\sin 2x| \text{ 的週期為 } \frac{\pi}{2}$$

$$(2) y = 3\sin x \text{ 的週期為 } 2\pi$$

$$(3) \because y = \cos x \text{ 的週期為 } 2\pi$$

$$\therefore y = \cos 2x \text{ 的週期為 } \frac{2\pi}{2} = \pi$$

$$(4) y = 4\cos x \text{ 的週期為 } 2\pi$$

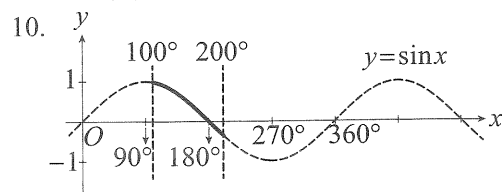
由(1)、(2)、(3)、(4)可知, 選項(C)是錯誤的

$$9. \because -1 \leq \sin \theta \leq 1$$

$$\therefore \text{港口 A: } \begin{cases} \text{滿潮: } 4 \times 1 + 16 = 20 \\ \text{乾潮: } 4 \times (-1) + 16 = 12 \end{cases}$$

$$\text{且港口 B: } \begin{cases} \text{滿潮: } -5 \times (-1) + 17 = 22 \\ \text{乾潮: } -5 \times 1 + 17 = 12 \end{cases}$$

故選(A)



當 θ 由 50° 增加到 100°

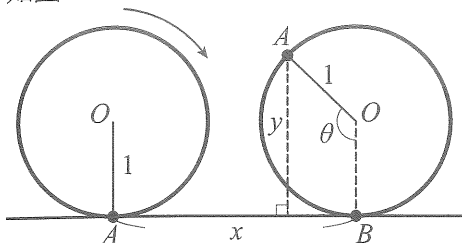
得 2θ 由 100° 增加到 200°

由正弦函數的圖形知

$$\sin(2\theta - 360^\circ) = \sin 2\theta \text{ 由正遞減到負}$$

故選(A)

11. 如圖

當輪子中心 O 前進 x 單位時扇形 OAB 所對應的弧長為 $1 \times \theta = \theta$,則 $y = f(x)$ 的對應如下:(1) 當 $x = \theta = 0$ 時, $y = 0$ (2) 當 $x = \theta = \frac{\pi}{2}$ 時, $y = 1$ (3) 當 $x = \theta = \pi$ 時, $y = 2$ (4) 當 $x = \theta = \frac{3\pi}{2}$ 時, $y = 1$ (5) 當 $x = \theta = 2\pi$ 時, $y = 0$ 故 $y = f(x)$ 的圖形通過 $(0,0)$ 、 $(\frac{\pi}{2},1)$ 、 $(\pi,2)$ 、 $(\frac{3\pi}{2},1)$ 、 $(2\pi,0)$ 因此, $y = f(x)$ 的圖形為選項(D)12.(1) $\sin 155^\circ = \sin(180^\circ - 25^\circ) = \sin 25^\circ$

$\because$ 在第一象限內, $\sin x$ 為遞增函數,
而 $25^\circ < 55^\circ$

 $\therefore \sin 25^\circ < \sin 55^\circ$,則 $\sin 155^\circ - \sin 55^\circ = \sin 25^\circ - \sin 55^\circ < 0$ (2) $\cos 222^\circ = \cos(180^\circ + 42^\circ) = -\cos 42^\circ$ $\cos 122^\circ = \cos(180^\circ - 58^\circ) = -\cos 58^\circ$

$\because$ 在第一象限內, $\cos x$ 為遞減函數,
而 $42^\circ < 58^\circ$

 $\therefore \cos 42^\circ > \cos 58^\circ \Rightarrow -\cos 42^\circ < -\cos 58^\circ$

則 $\cos 222^\circ - \cos 122^\circ$
 $= -\cos 42^\circ - (-\cos 58^\circ) < 0$

由(1)和(2)可知,

點 $(\sin 155^\circ - \sin 55^\circ, \cos 222^\circ - \cos 122^\circ)$

在第三象限

$$\begin{aligned} 13.(1) \because ab : bc : ca &= \frac{ab}{abc} : \frac{bc}{abc} : \frac{ca}{abc} \\ &= \frac{1}{c} : \frac{1}{a} : \frac{1}{b} \end{aligned}$$

且 $ab : bc : ca = 3 : 4 : 6$

$$\therefore \frac{1}{c} : \frac{1}{a} : \frac{1}{b} = 3 : 4 : 6$$

$$\text{則 } c : a : b = \frac{1}{3} : \frac{1}{4} : \frac{1}{6}$$

$$= \frac{1}{3} \times 12 : \frac{1}{4} \times 12 : \frac{1}{6} \times 12 = 4 : 3 : 2$$

$$\Rightarrow a : b : c = 3 : 2 : 4$$

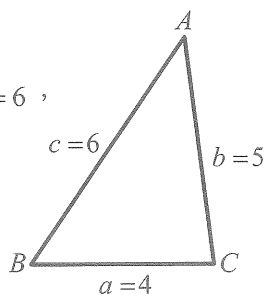
(2) 由正弦定理可知,

$$\sin A : \sin B : \sin C = a : b : c = 3 : 2 : 4$$

14. [解1]

設 $\triangle ABC$ 的

$$\overline{BC} = 4, \overline{AC} = 5, \overline{AB} = 6,$$

則 $a = 4, b = 5, c = 6$ 

(1) 由餘弦定理

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5^2 + 6^2 - 4^2}{2 \times 5 \times 6} = \frac{45}{60} = \frac{3}{4}$$

(2) $\therefore \sin^2 A + \cos^2 A = 1$

$$\therefore \sin^2 A = 1 - \cos^2 A = 1 - \left(\frac{3}{4}\right)^2 = 1 - \frac{9}{16} = \frac{7}{16}$$

$$\Rightarrow \sin A = \pm \sqrt{\frac{7}{16}} = \pm \frac{\sqrt{7}}{4}$$

而當 $0^\circ < \angle A < 180^\circ$ 時, $\sin A$ 恆正,

$$\text{則 } \sin A = \frac{\sqrt{7}}{4}$$

(3) 設 $\triangle ABC$ 的外接圓半徑為 R

由正弦定理

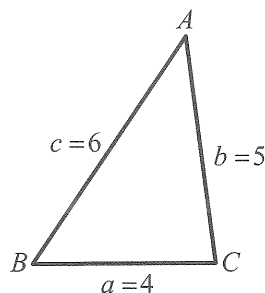
$$\frac{a}{\sin A} = 2R \Rightarrow \frac{4}{\frac{\sqrt{7}}{4}} = 2R \Rightarrow 2R = \frac{16}{\sqrt{7}}$$

因此 $\triangle ABC$ 的外接圓直徑 $2R = \frac{16}{\sqrt{7}}$

[解2]

設 $\triangle ABC$ 的 $\overline{BC} = 4, \overline{AC} = 5, \overline{AB} = 6,$ 則 $a = 4, b = 5, c = 6$

$$(1) \text{ 令 } s = \frac{1}{2}(a+b+c) = \frac{1}{2} \times (4+5+6) = \frac{15}{2}$$



由海龍公式

 $\triangle ABC$ 面積

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$\begin{aligned}
 &= \sqrt{\frac{15}{2} \times \left(\frac{15}{2} - 4\right) \times \left(\frac{15}{2} - 5\right) \times \left(\frac{15}{2} - 6\right)} \\
 &= \sqrt{\frac{15}{2} \times \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2}} = \frac{15}{4} \sqrt{7}
 \end{aligned}$$

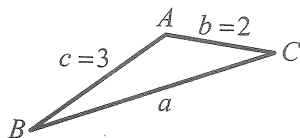
(2) 設 $\triangle ABC$ 的外接圓半徑為 R ,

$$\text{則 } \triangle ABC \text{ 面積} = \frac{abc}{4R} = \frac{4 \times 5 \times 6}{4R} = \frac{30}{R}$$

$$\text{由(1)、(2)可知, } \frac{30}{R} = \frac{15}{4} \sqrt{7} \Rightarrow R = \frac{8}{\sqrt{7}},$$

$$\text{因此 } \triangle ABC \text{ 的外接圓直徑 } 2R = 2 \times \frac{8}{\sqrt{7}} = \frac{16}{\sqrt{7}}$$

15. 令 $b = \overline{AC} = 2$, $c = \overline{AB} = 3$



(1) $\triangle ABC$ 的面積

$$= \frac{1}{2} bc \sin A = \frac{1}{2} \times 2 \times 3 \times \sin A = 3 \sin A$$

$$\text{則 } 3 \sin A = \frac{3\sqrt{2}}{2} \Rightarrow \sin A = \frac{\sqrt{2}}{2}$$

$$\therefore \sin 45^\circ = \sin 135^\circ = \frac{\sqrt{2}}{2}, \text{ 而 } \angle A \text{ 是鈍角}$$

$$\therefore \angle A = 135^\circ$$

(2) 由餘弦定理:

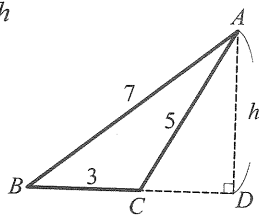
$$\begin{aligned}
 a^2 &= b^2 + c^2 - 2bc \cos A \\
 &= 2^2 + 3^2 - 2 \times 2 \times 3 \times \cos 135^\circ \\
 &= 4 + 9 - 2 \times 2 \times 3 \times \left(-\frac{\sqrt{2}}{2}\right) = 13 + 6\sqrt{2}
 \end{aligned}$$

16. (1) 設 $\overline{BC}$ 邊上的高 $\overline{AD} = h$

則 $\triangle ABC$ 面積

$$= \frac{1}{2} \times \overline{BC} \times h$$

$$= \frac{1}{2} \times 3 \times h = \frac{3}{2} h$$



(2) 令 $\overline{BC} = a$, $\overline{AC} = b$, $\overline{AB} = c$

$$\text{則 } a = 3, b = 5, c = 7$$

$$\text{令 } s = \frac{1}{2}(a+b+c) = \frac{1}{2} \times (3+5+7) = \frac{15}{2}$$

由海龍公式:

$\triangle ABC$ 面積

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{\frac{15}{2} \times \left(\frac{15}{2} - 3\right) \times \left(\frac{15}{2} - 5\right) \times \left(\frac{15}{2} - 7\right)}$$

$$= \sqrt{\frac{15}{2} \times \frac{9}{2} \times \frac{5}{2} \times \frac{1}{2}} = \frac{15}{4} \sqrt{3}$$

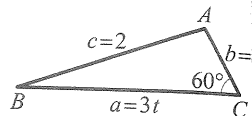
$$\text{由(1)和(2), } \frac{3}{2} h = \frac{15}{4} \sqrt{3} \Rightarrow h = \frac{5\sqrt{3}}{2}$$

因此 $\overline{AD} = \frac{5\sqrt{3}}{2}$, 故選(D)

17. $\therefore \overline{AC} : \overline{BC} = 1 : 3$

$\therefore$ 設 $\overline{AC} = t$,

$$\overline{BC} = 3t \quad (t > 0)$$



即 $\triangle ABC$ 的 $a = 3t$, $b = t$, $c = 2$

(1) 由餘弦定理 $c^2 = a^2 + b^2 - 2ab \cos C$

$$\text{則 } 2^2 = (3t)^2 + t^2 - 2 \times 3t \times t \times \cos 60^\circ$$

$$\Rightarrow 4 = 9t^2 + t^2 - 6t^2 \times \frac{1}{2} \Rightarrow 7t^2 = 4 \Rightarrow t^2 = \frac{4}{7}$$

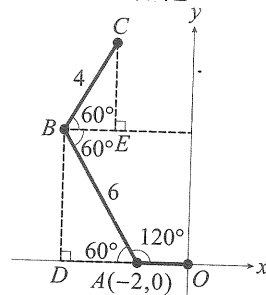
(2) $\triangle ABC$ 的面積 $= \frac{1}{2} ab \sin C$

$$= \frac{1}{2} \times 3t \times t \times \sin 60^\circ = \frac{3}{2} t^2 \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4} t^2$$

$$= \frac{3\sqrt{3}}{4} \times \frac{4}{7} = \frac{3\sqrt{3}}{7}$$

18. [解1]

將圖形坐標化:



令 O 為原點, A 點為 $(-2,0)$

(1) 從 B 點對 x 軸作垂直線交於 D 點

$$\therefore \angle OAB = 120^\circ$$

$$\therefore \angle BAD = 180^\circ - 120^\circ = 60^\circ$$

在 $\triangle ABD$ 之中

$$\overline{AD} = \overline{AB} \times \cos 60^\circ = 6 \times \frac{1}{2} = 3$$

$$\overline{BD} = \overline{AB} \times \sin 60^\circ = 6 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$$

$$\text{則 } B \text{ 點坐標為 } (-2-3, 0+3\sqrt{3}) = (-5, 3\sqrt{3})$$

(2) 過 B 點作水平線平行 x 軸,

從 C 點對水平線作垂直線交於 E 點

$$\therefore \angle EBA = \angle BAD = 60^\circ \quad (\text{內錯角相等})$$

$$\therefore \angle CBE = 120^\circ - 60^\circ = 60^\circ$$

在 $\triangle BCE$ 之中

$$\overline{BE} = \overline{BC} \times \cos 60^\circ = 4 \times \frac{1}{2} = 2$$

$$\overline{CE} = \overline{BC} \times \sin 60^\circ = 4 \times \frac{\sqrt{3}}{2} = 2\sqrt{3}$$

則C點坐標為

$$(-5 + 2, 3\sqrt{3} + 2\sqrt{3}) = (-3, 5\sqrt{3})$$

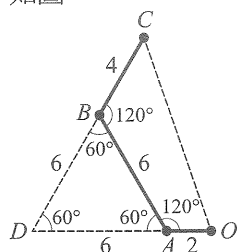
$$\begin{aligned}\text{故 } \overline{OC} &= \sqrt{(-3-0)^2 + (5\sqrt{3}-0)^2} \\ &= \sqrt{9+75} = \sqrt{84} = 2\sqrt{21}\end{aligned}$$

因此C地到O地的直線距離為 $2\sqrt{21}$ 公里，

故選(B)

〔解2〕

(1) 如圖



連接線段 $\overline{OC}$ ，延長 $\overline{BC}$ 與 $\overline{OA}$ 交於 D 點

$$\because \angle OAB = 120^\circ$$

$$\therefore \angle BAD = 180^\circ - 120^\circ = 60^\circ$$

$$\because \angle ABC = 120^\circ$$

$$\therefore \angle ABD = 180^\circ - 120^\circ = 60^\circ$$

$$\begin{aligned}\text{而 } \angle D &= 180^\circ - \angle BAD - \angle ABD \\ &= 180^\circ - 60^\circ - 60^\circ = 60^\circ\end{aligned}$$

則 $\triangle ABD$ 是正三角形

$$\Rightarrow \overline{AD} = \overline{BD} = \overline{AB} = 6$$

$$(2) \overline{OD} = \overline{OA} + \overline{AD} = 2 + 6 = 8$$

$$\overline{CD} = \overline{CB} + \overline{BD} = 4 + 6 = 10$$

在 $\triangle OCD$ 之中，由餘弦定理

$$\overline{OC}^2 = \overline{OD}^2 + \overline{CD}^2 - 2 \times \overline{OD} \times \overline{CD} \times \cos D$$

$$= 8^2 + 10^2 - 2 \times 8 \times 10 \times \cos 60^\circ$$

$$= 64 + 100 - 2 \times 8 \times 10 \times \frac{1}{2} = 84$$

$$\Rightarrow \overline{OC} = \sqrt{84} = 2\sqrt{21}$$

因此C地到O地的距離為 $2\sqrt{21}$ 公里

故選(B)

19.(1) $\because \sin x$ 的週期為 2π

$$\text{而 } f(x) = 40 \sin \left(ax + \frac{\pi}{2} \right) + 70 \text{ 且 } a > 0$$

$$\therefore f(x) \text{ 的週期為 } \frac{2\pi}{a} \text{ (秒)}$$

(2) $\because$ 風車逆時針等速轉一圈需時 4 秒

$$\therefore f(x) \text{ 的週期為 } 4 \text{ (秒)}$$

$$\text{由(1)和(2), } \frac{2\pi}{a} = 4 \Rightarrow a = \frac{\pi}{2}, \text{ 故選(B)}$$